

THE PROJECT CONTENTS:

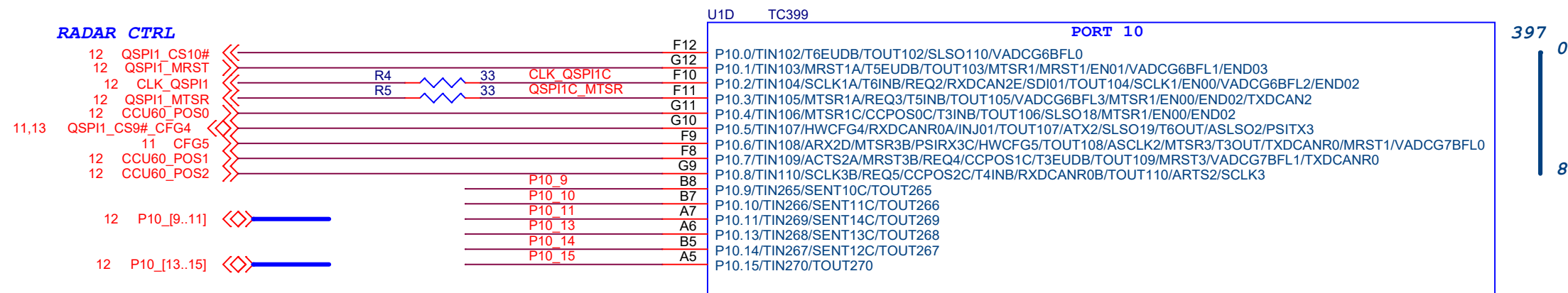
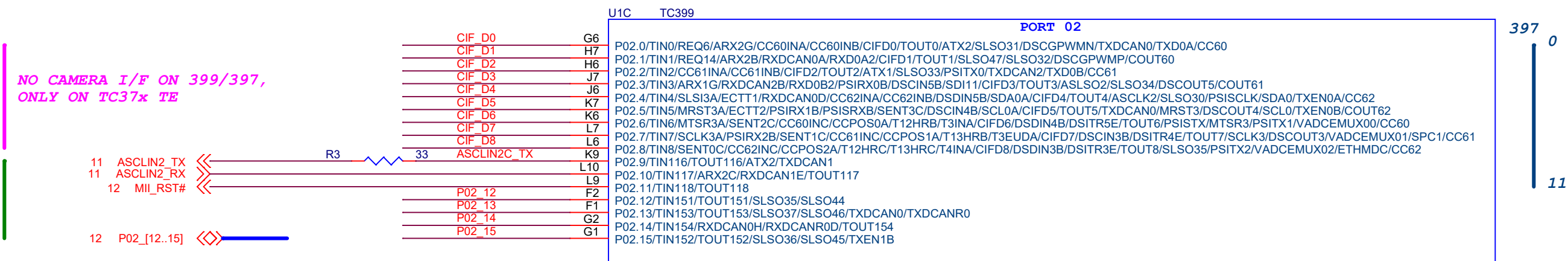
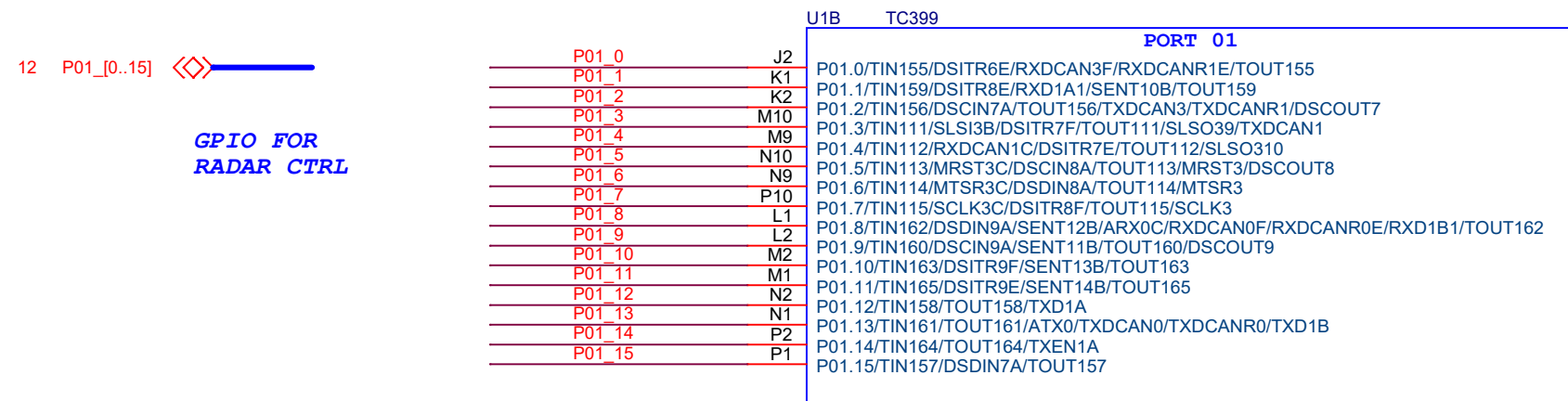
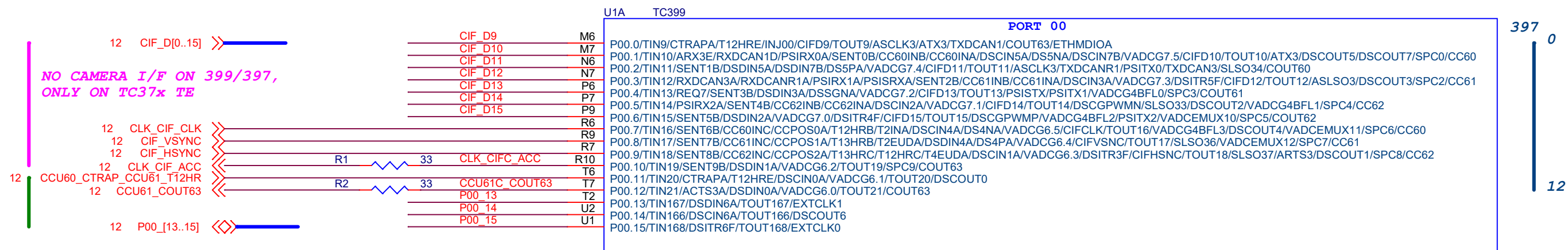
- 1. Title Page
- 2. Block Diagram
- 3. MCU Ports 00, 01, 02, 10
- 4. MCU Ports 11, 12, 13, 14, 15
- 5. MCU Ports 20, 21, 22, 23, 24 and HSCT/SDRAM Mux
- 6. MCU Ports 25, 30, 31, 32, 33, 34
- 7. MCU Port 40 and Analog Inputs 0-71
- 8. MCU Clock, System and Secondary DC/DC
- 9. MCU Power Connections and Decoupling
- 10. Power, Temperature Indication and Monitoring
- 11. Module System Connections
- 12. Module Interposer Connector 1
- 13. Module Interposer Connector 2
- 14. Mechanical Section and Notes

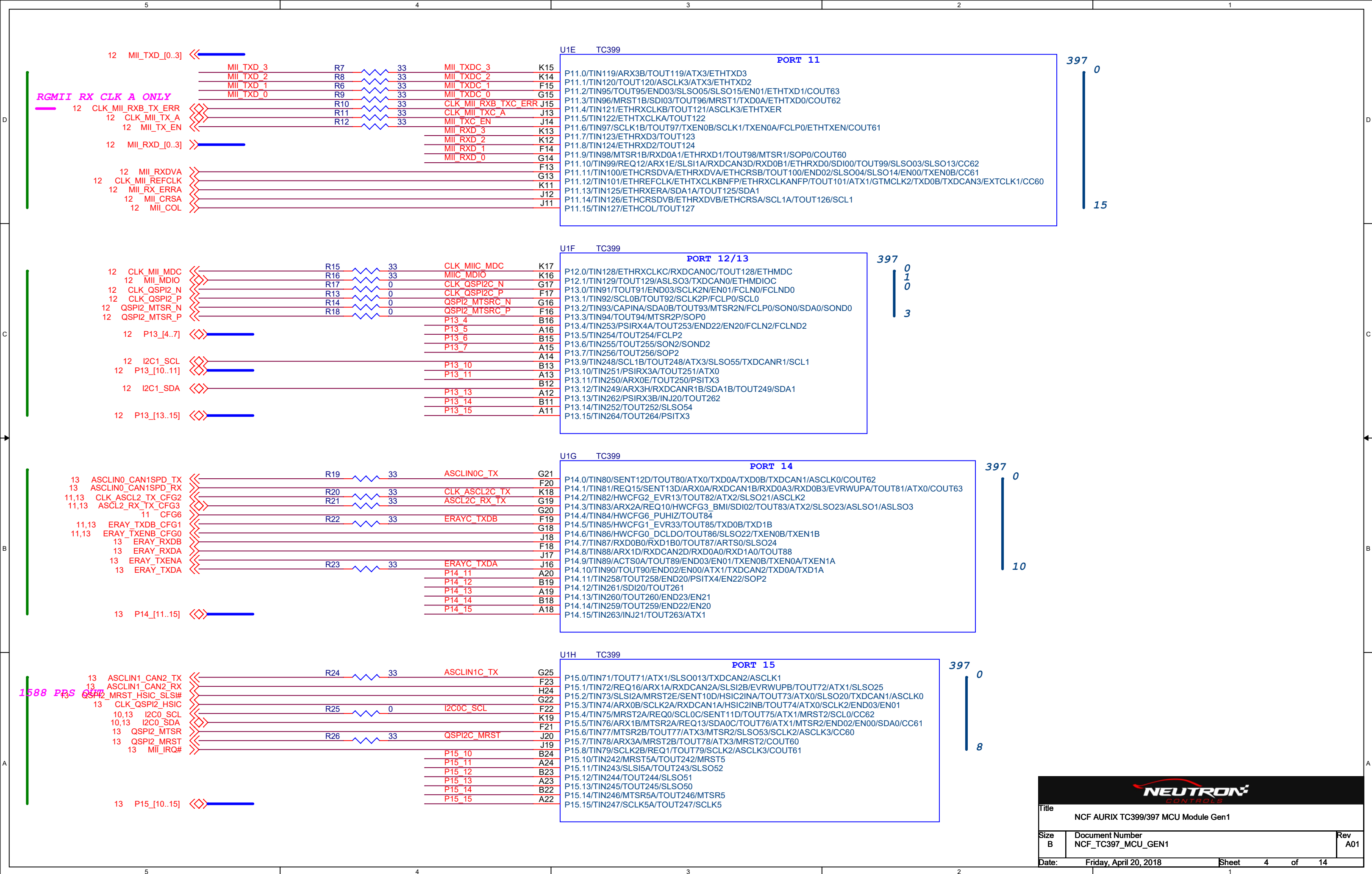
REVISION HISTORY:

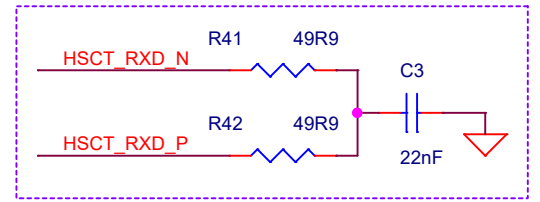
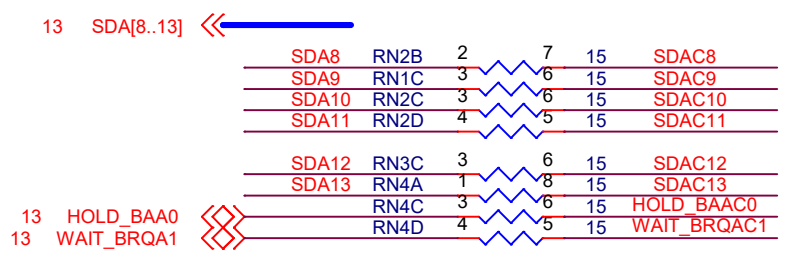
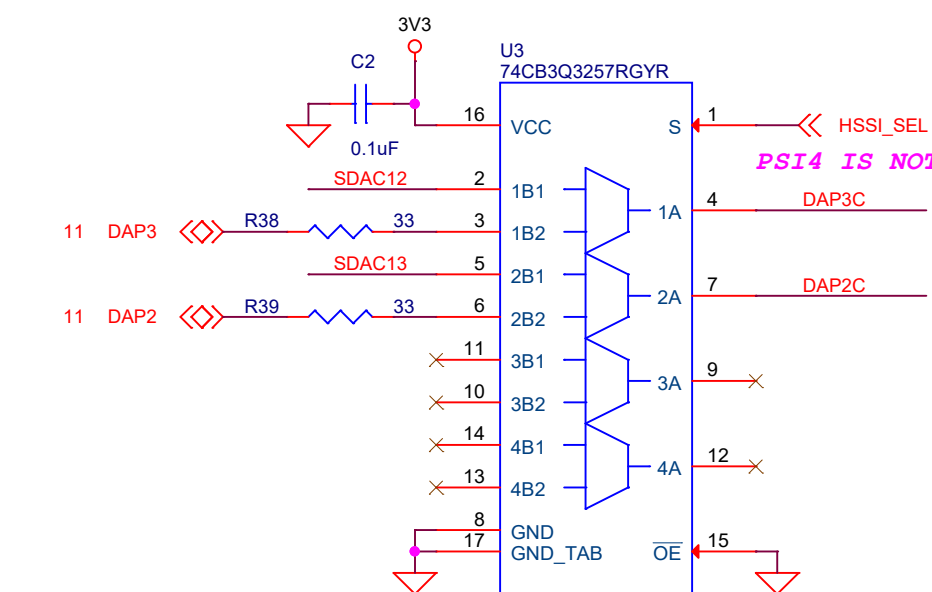
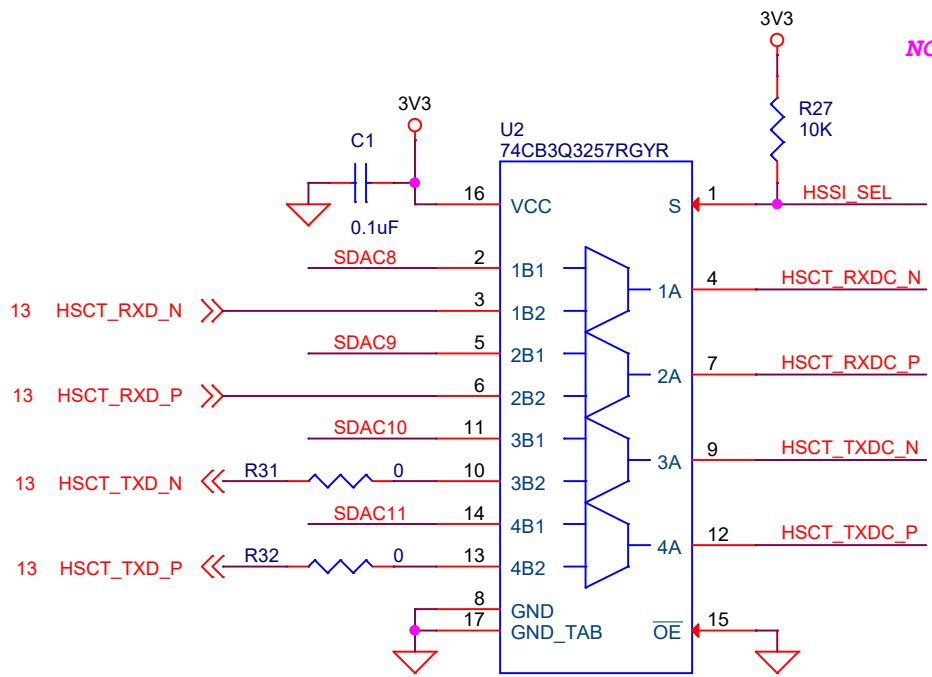
REV.	DATE	CHANGES	RESP.
A01	2017-01-15	INITIAL RELEASE (pre release)	A.G.



		
Title		
NCF AURIX TC399/397 MCU Module Gen1		
Size	Document Number	Rev
B	NCF_TC397_MCU_GEN1	A01
Date:	Wednesday, April 18, 2018	Sheet 2 of 14

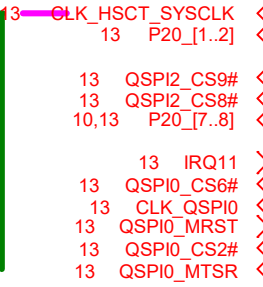






VERY SHORT (<1MM) STUBS
PLACE AT U1 PINS

NOT HSCT CLK IN

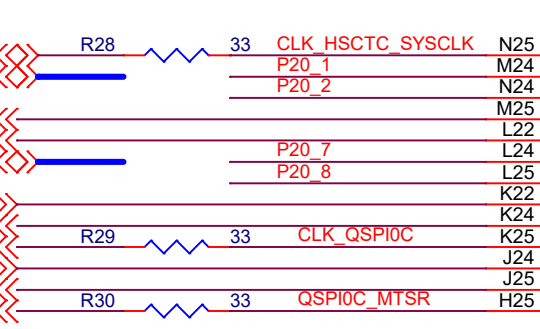


HOLD AND WAIT NOT AVAILABLE

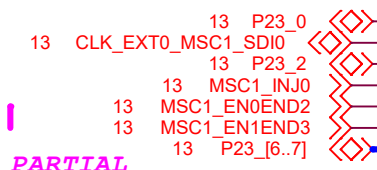
DIFF RECEIVER
TERMINATION
NEEDED

PSI4 IS NOT AVAILABLE !

PARTIAL

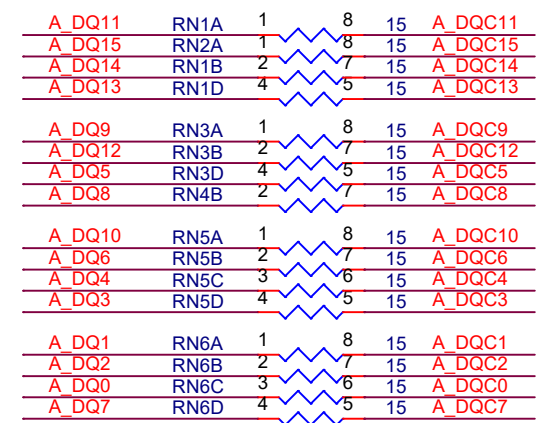


AS SHORT AS
POSSIBLE, N/ P
LENGTH MATCHED

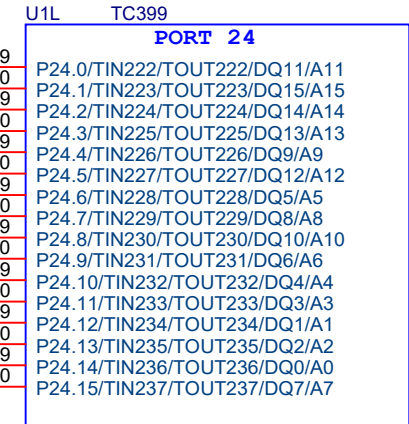
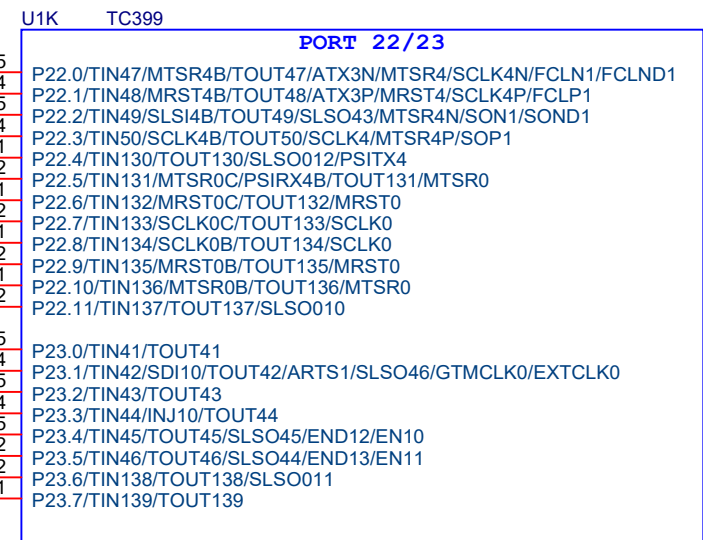
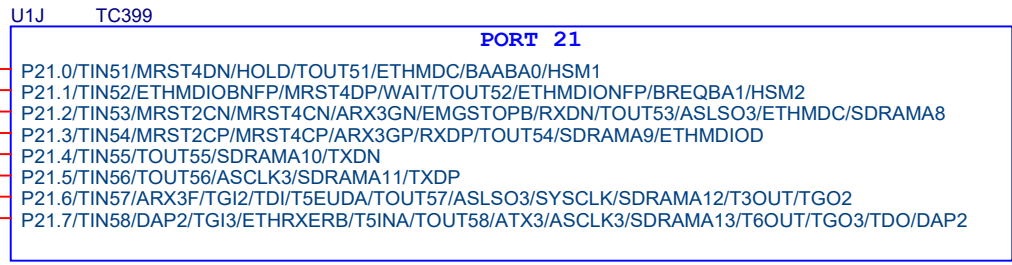
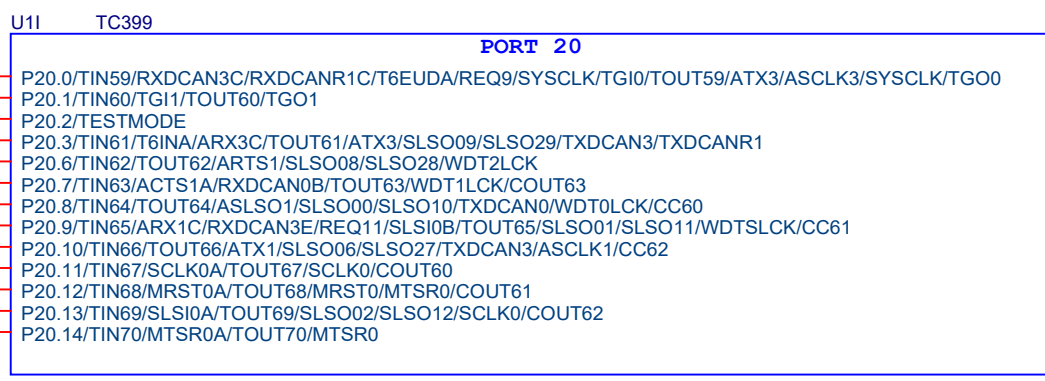
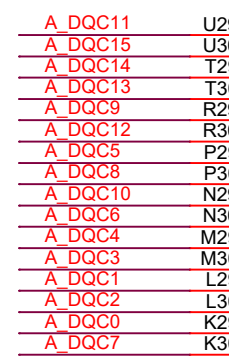


LENGTH MATCH
TO +/-1MM
(WITH X-NETS)

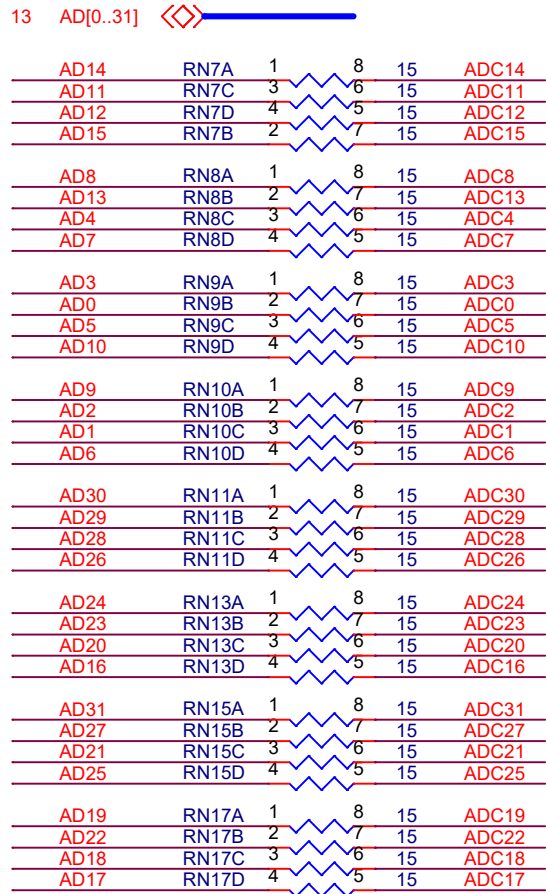
SWAP RXN AS NEEDED



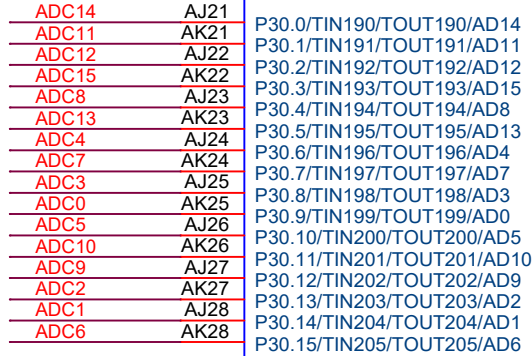
ADDRESS ONLY ?



397 - NO PINS

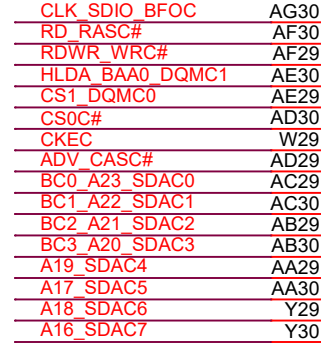


DATA ONLY ?



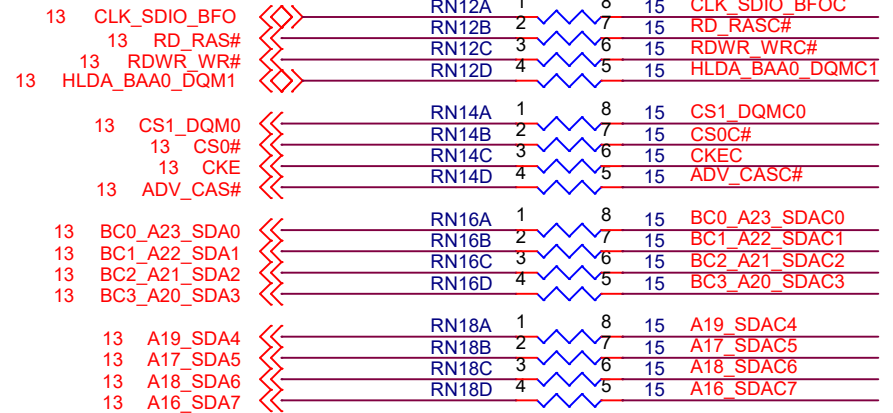
397 - NO PINS

PARTIAL COMP., REMAPPING



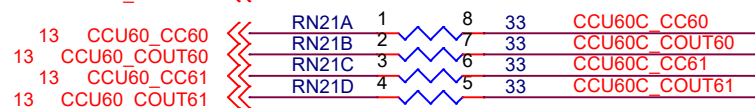
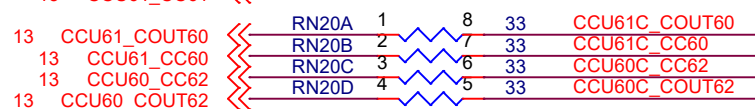
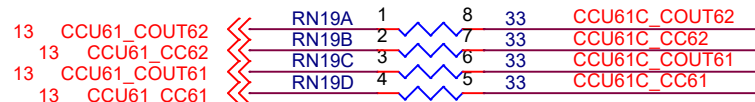
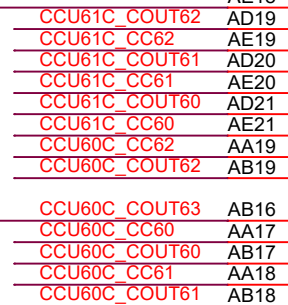
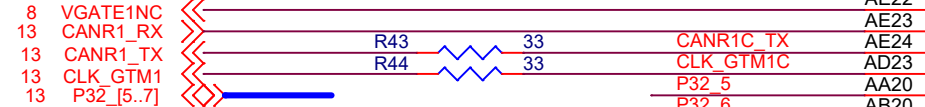
397 - NO PINS

13 CLK_BFI >>>



397 - NO PINS

PSI5 VS. PSI0



U1P TC399

PORT 32/33/34



397 0 2 7 0 15 1 5



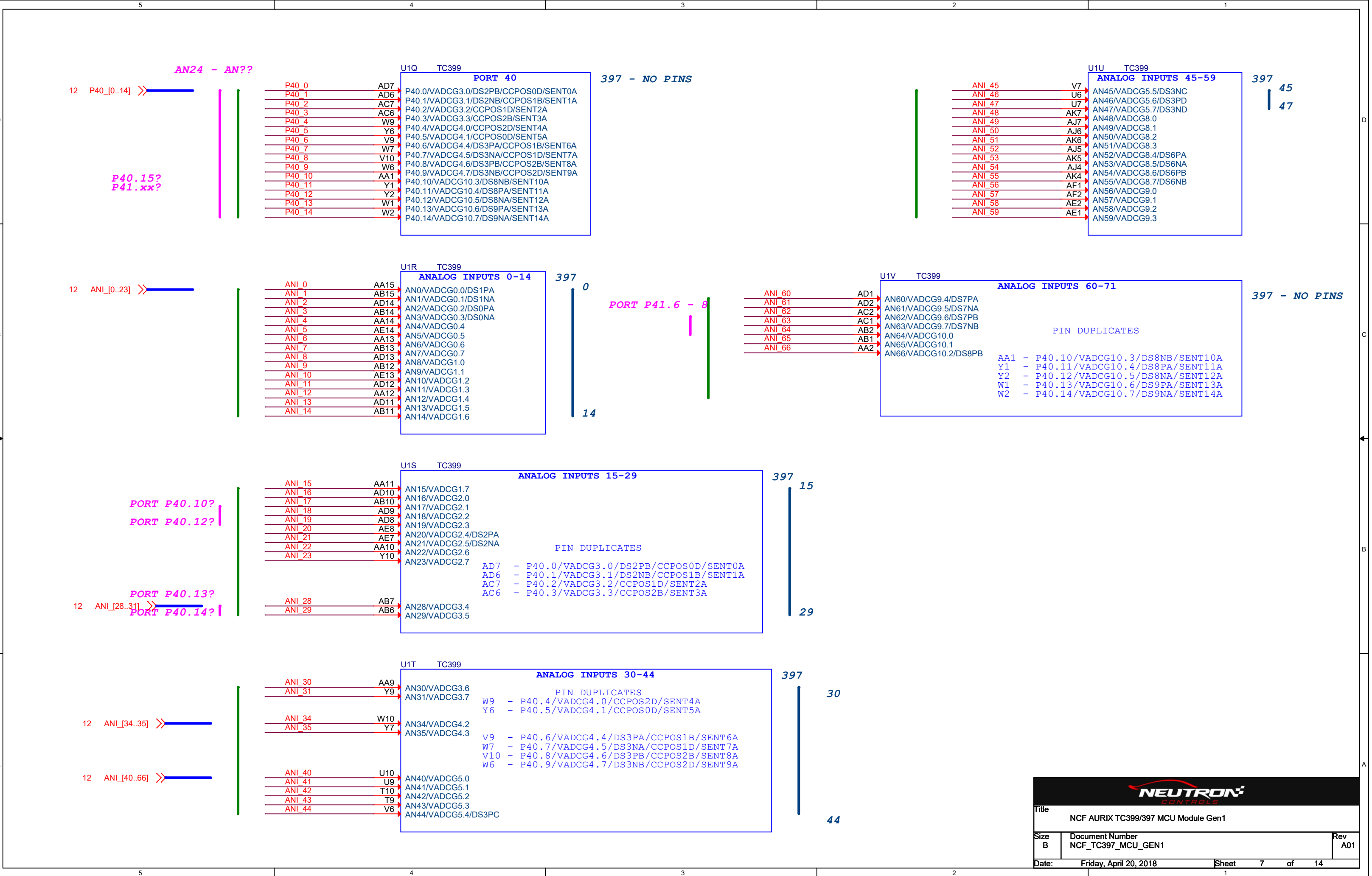
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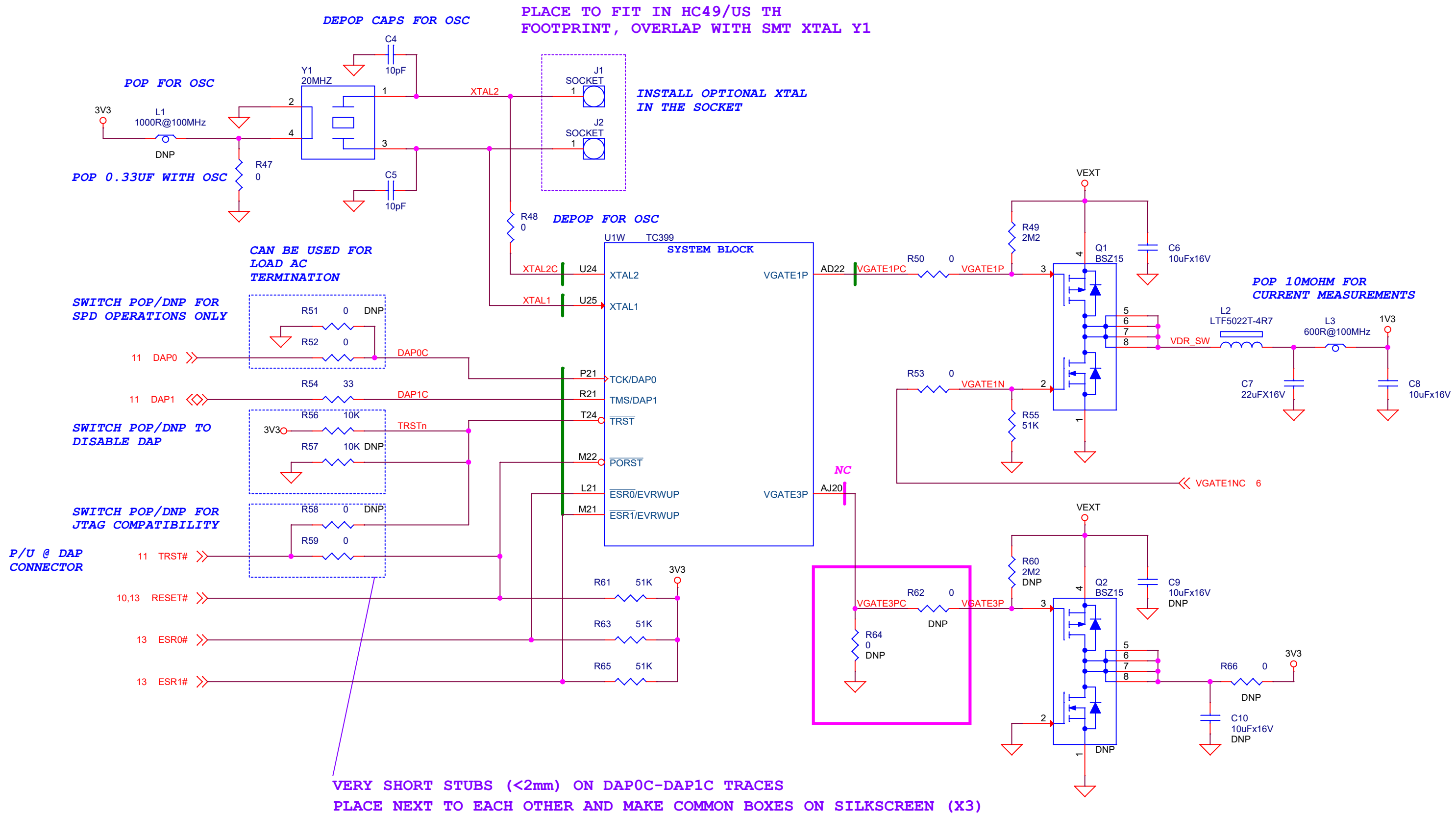
Size B Document Number NCF_TC397_MCU_GEN1

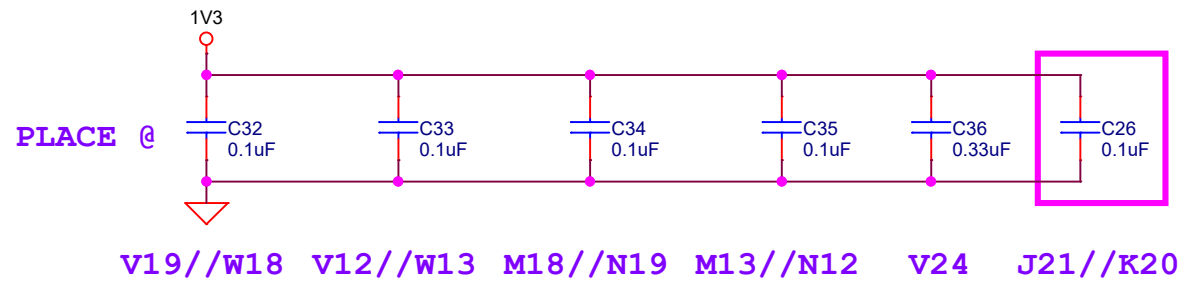
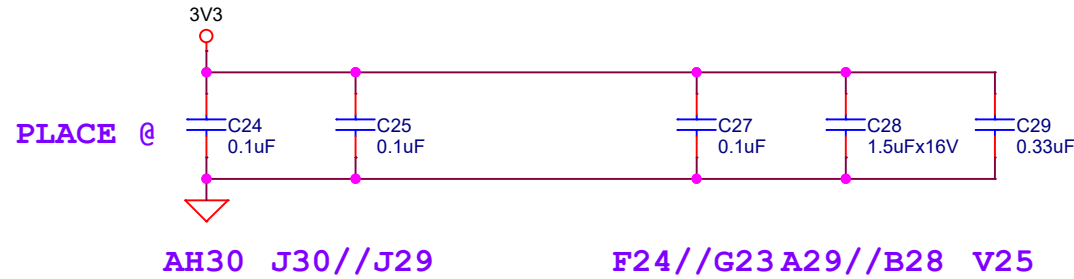
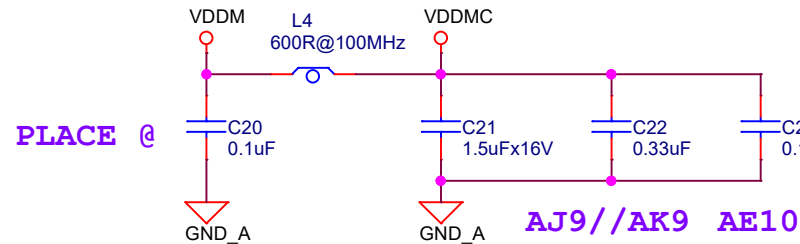
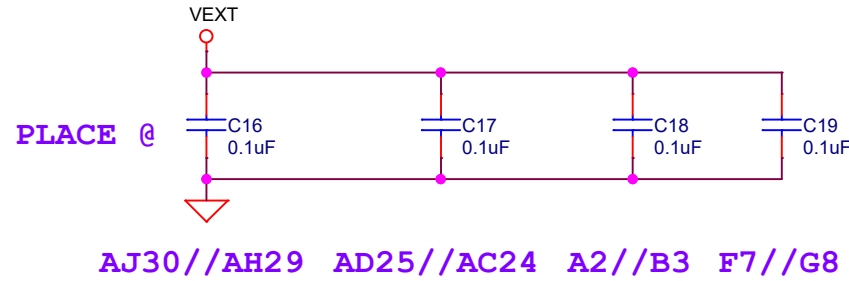
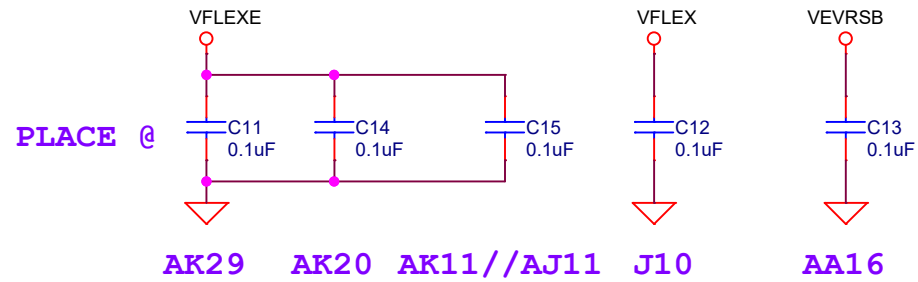
Rev A01

Date: Friday, April 20, 2018

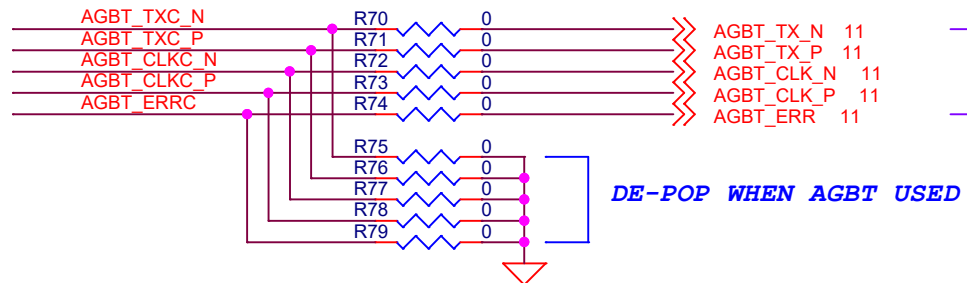
Sheet 6 of 14







2.5GBPS SIGNALS, PLACE WITH
VERY SHORT (<1MM) STUBS



EXTERNAL SUPPLY

SRAM SB SUPPLY

FLEX PADS SUPPLY

FLEX EBU PADS SUPPLY

EBU SUPPLY

ADC SUPPLY

VEXT

FLASH SUPPLY

EBU SUPPLY

CORE SUPPLY

ED SRAM SUPPLY

ED DAP SUPPLY

N/C, DAP CLK ED

VAREF3/GND3

LENGTH MATCHED
GROUP (0.25MM)

AN72-73

U1X TC399

POWER BLOCK

5V/3.3V

3.3V

1.3V

A2
B3
F7
G8
AC24
AD25
AH29
AJ30

VEVRSB

J10

VFLEX

VFLEX1

VFLEX2

VFLEX3

VFLEX4

VDDM1

VDDM2

VDDM3

VDDP3_1

VDDP3_2

VDDP3_3

VDDP3_4

VDDP3/OSC_LVDSH

VDDFL3_1

VDDFL3_2

VEBU1

VEBU2

VEBU3

VDD1

VDD2

VDD3

VDD4

VDD5

VDD6

VDD7/XTAL

VDD / VDDSB1

VDD / VDDSB2

VAREF1

VAREF2

VAGND1

VAREF2

VAGND2

NC/VDDPSB

NC_FE1

NC_FE2

NC_FE3

NC_FE4

NC_FE5

NC_FE6

NC_FE7

NC_FE8

NC_FE9

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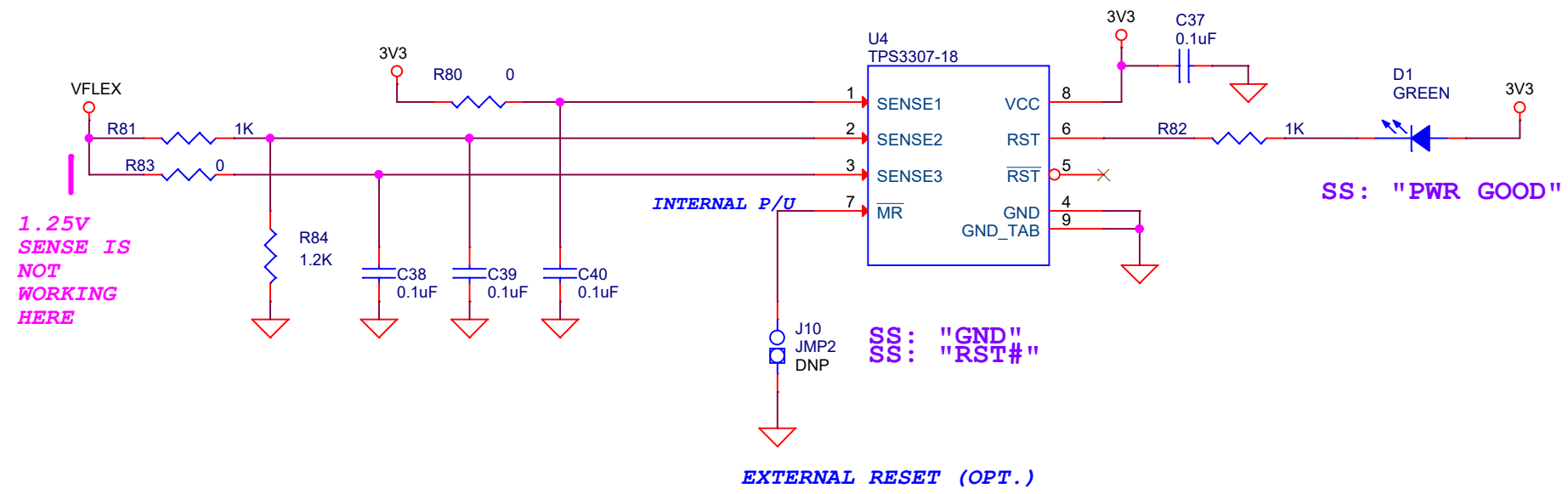
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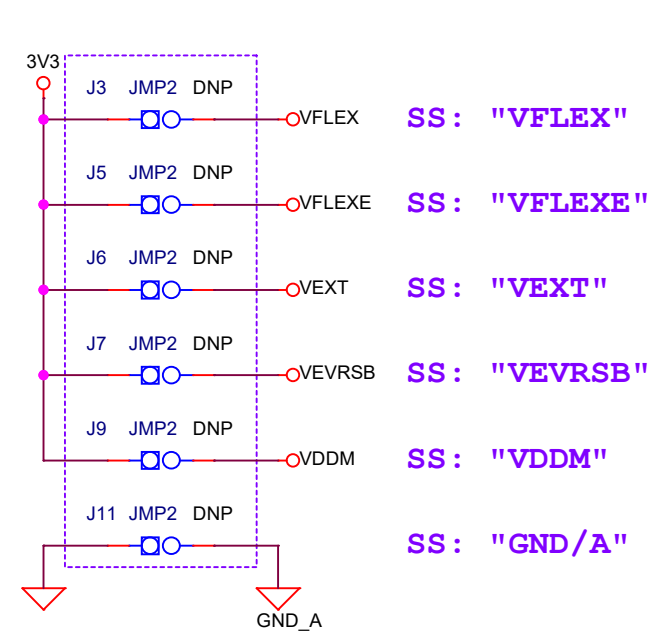
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NC_FE230

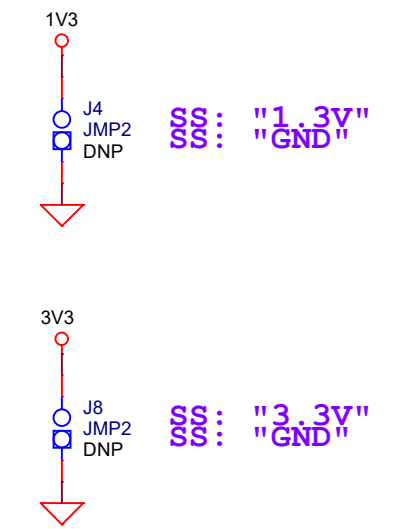
NC_FE231



POPULATE FOR "MODULE ONLY" POWER CONFIGURATION

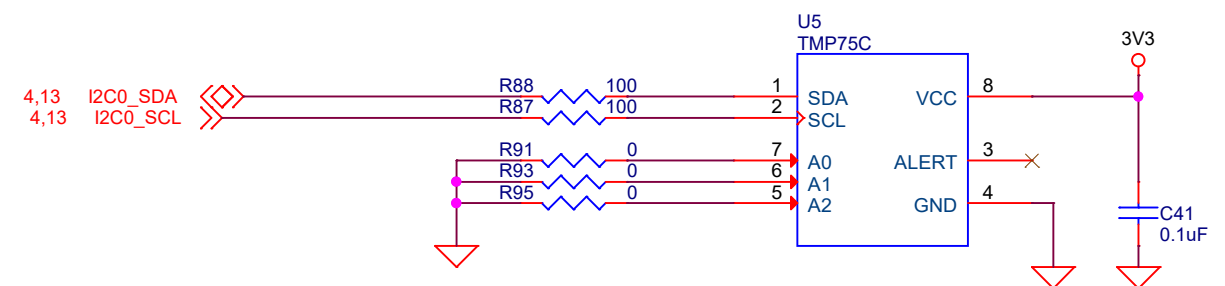


VOLTAGE MEASUREMENT POINTS



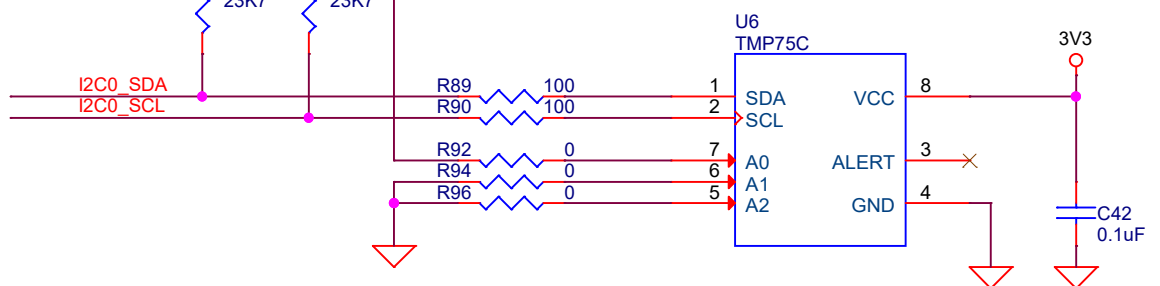
PLACE IN THE SAME BLOCK WITH 2MM PITCH

CPU TEMPERATURE SENSOR

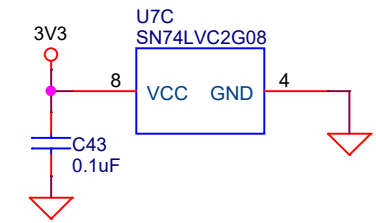
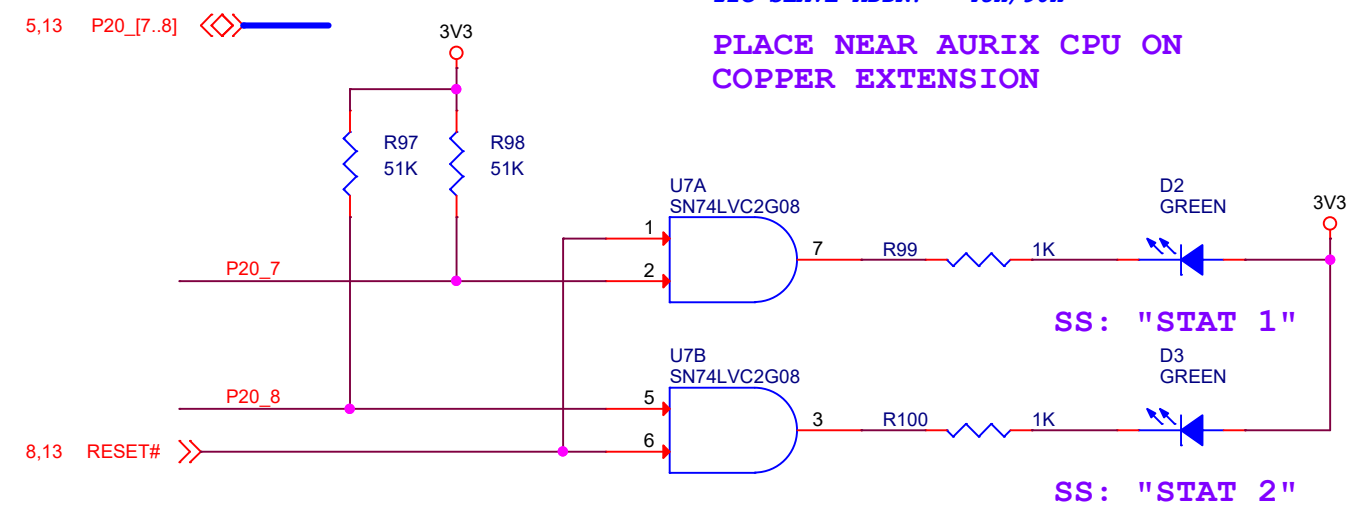


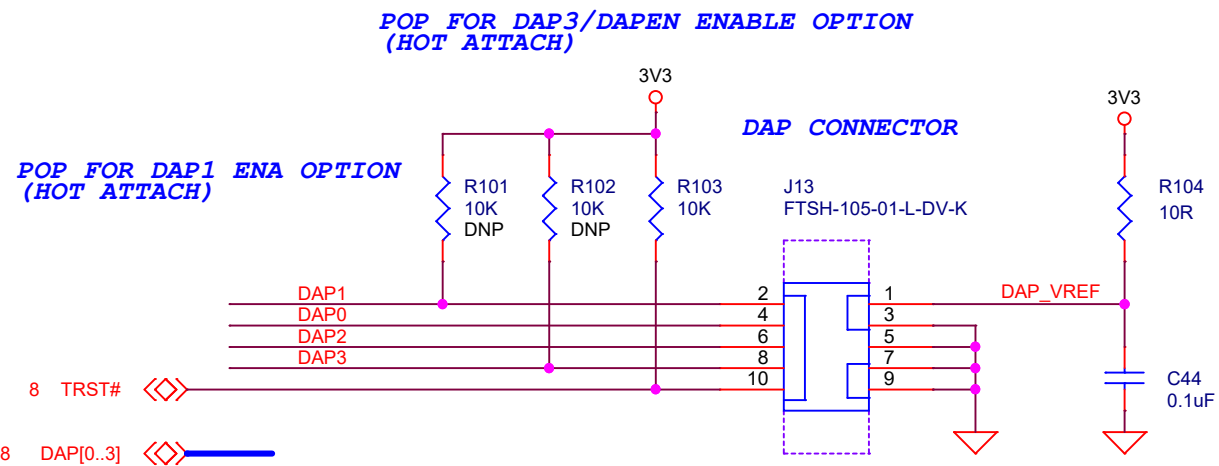
PLACE NEAR AURIX CPU ON COPPER EXTENSION

MOSFET TEMPERATURE SENSOR



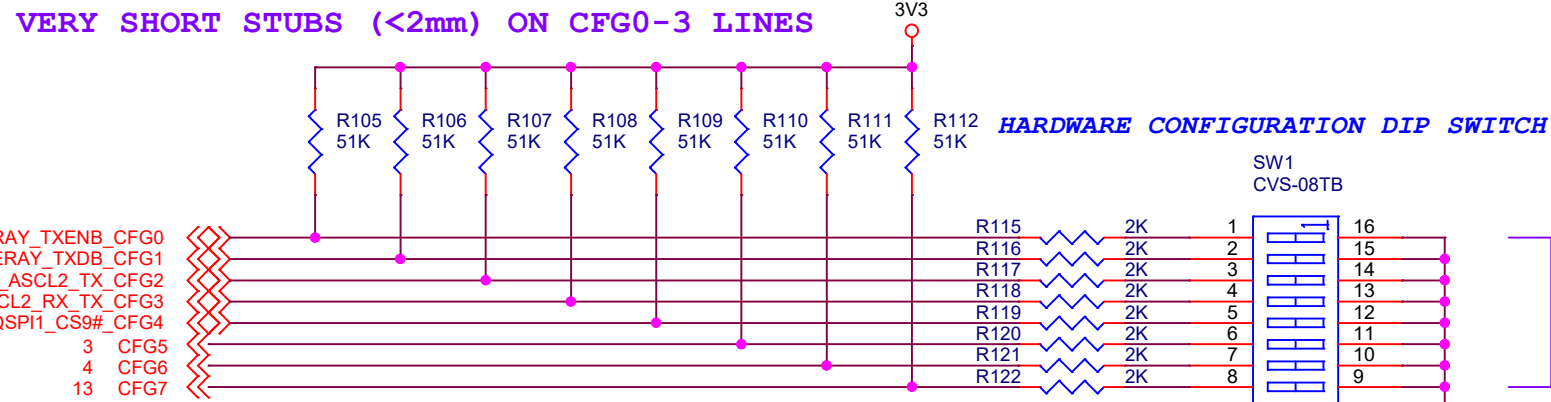
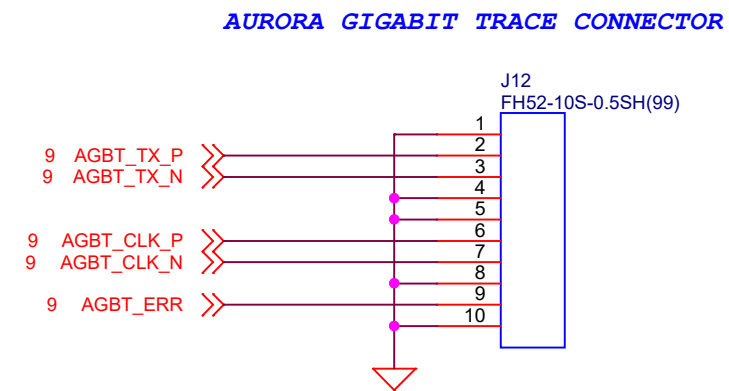
PLACE NEAR MOSFET Q1 ON COPPER EXTENSION





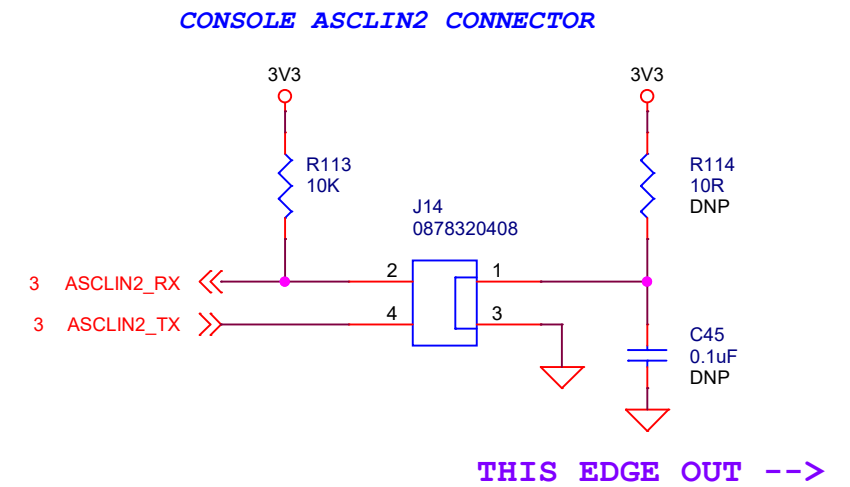
VERY SHORT STUBS (<2mm) ON DAP0-DAP3 TRACES
DAP0-DAP2 TRACES LENGTH MATCHED (<1mm)

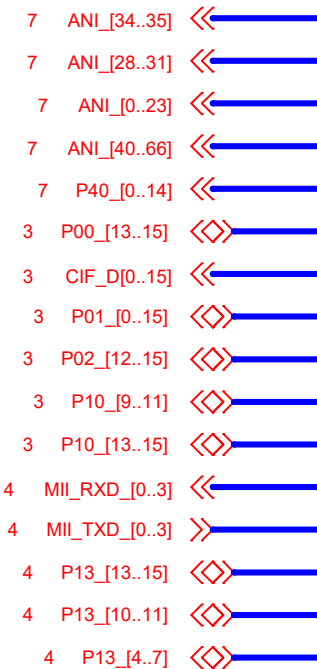
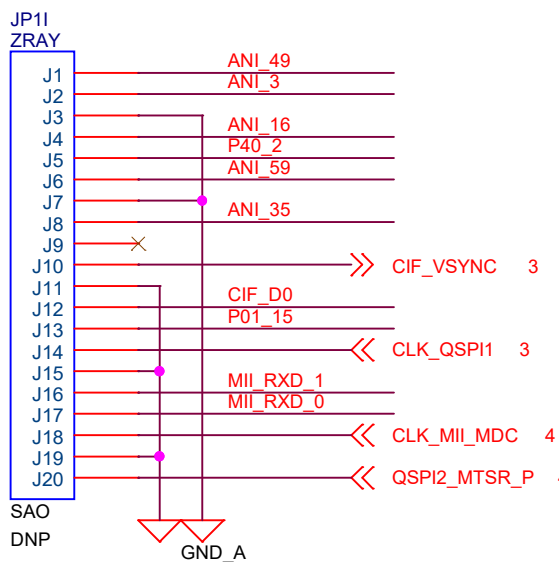
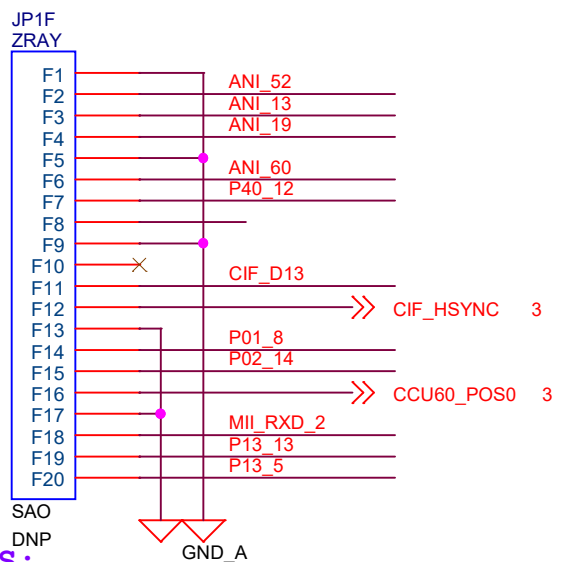
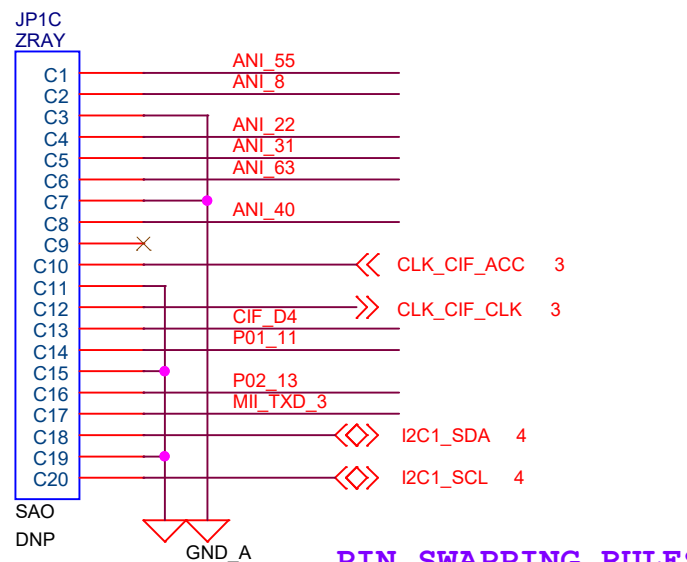
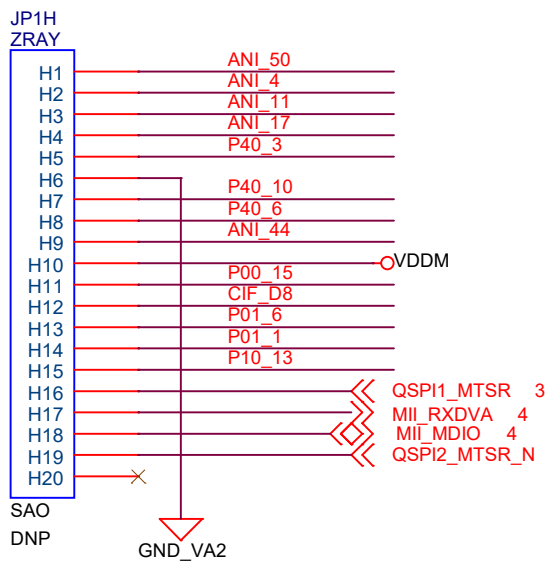
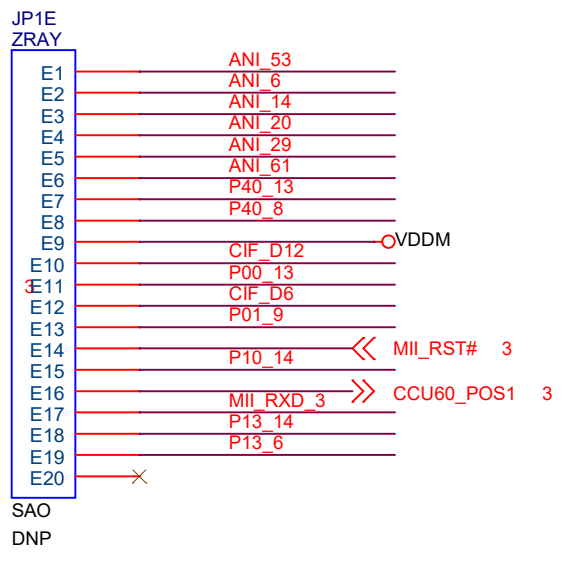
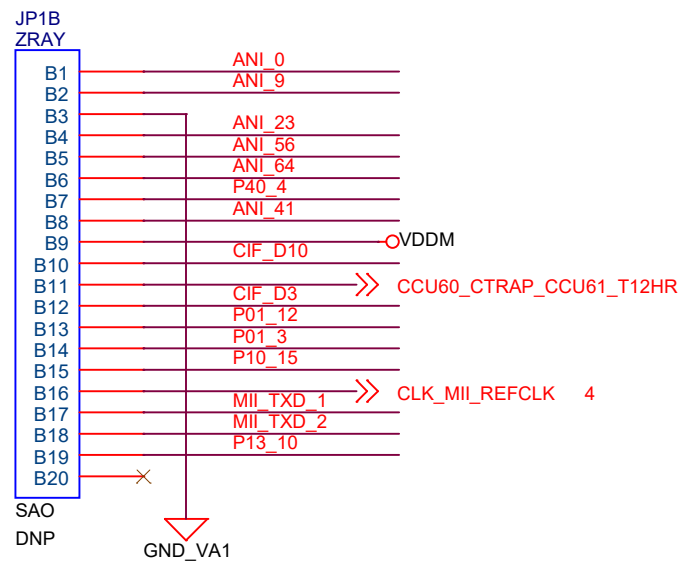
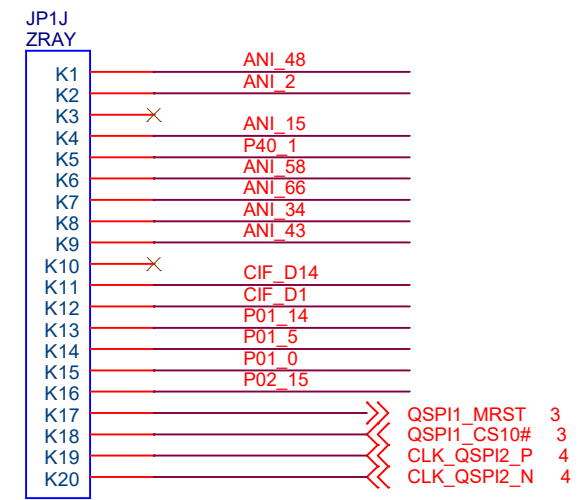
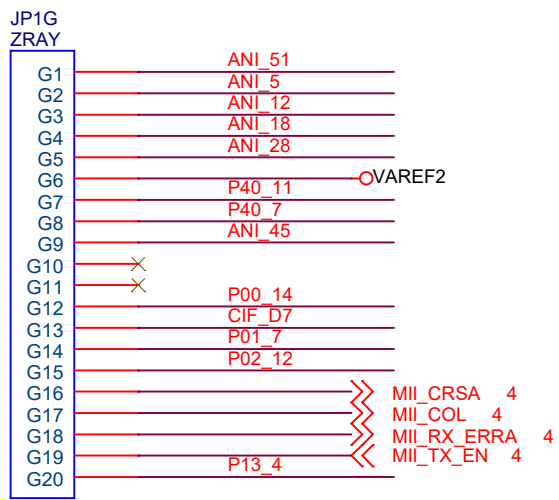
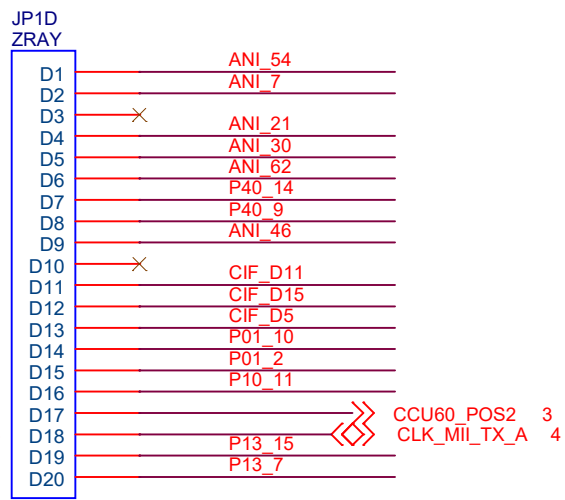
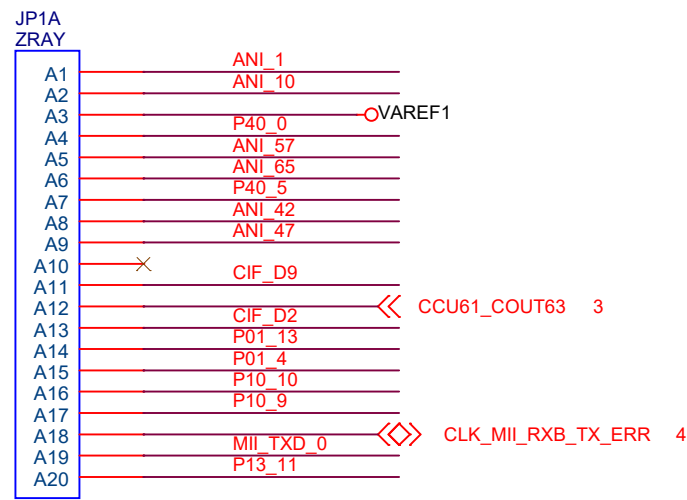
THIS EDGE OUT -->
LEAVE EXTRA 2.5mm ON EACH
SIDE FOR CABLE PLUG (>2mm H)



SS: " HWCFG0-7"

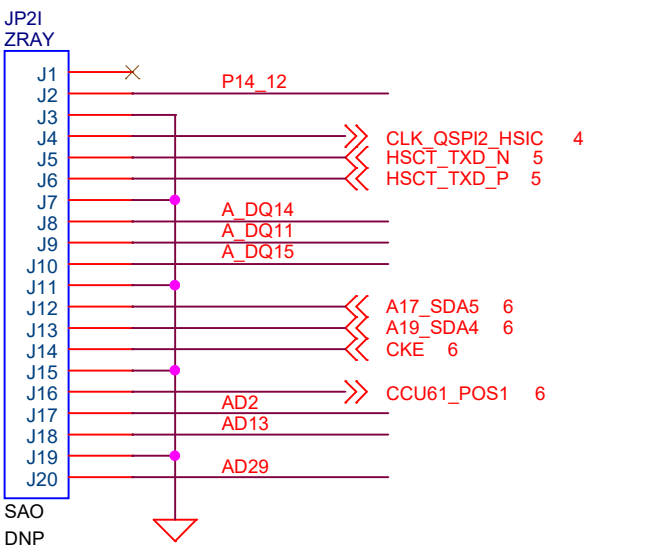
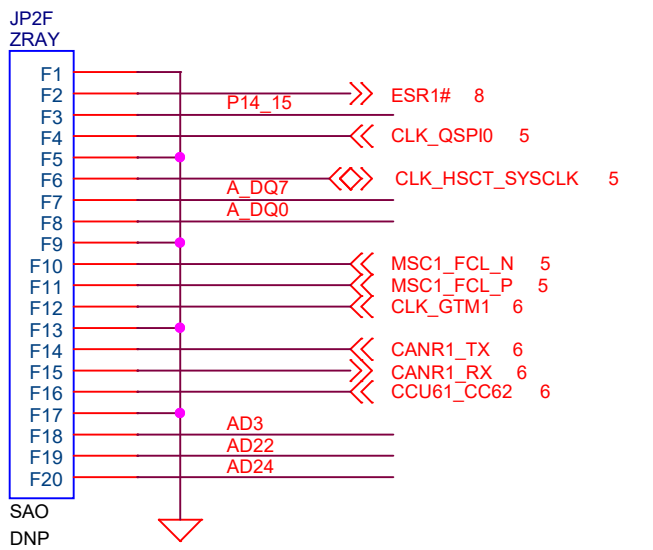
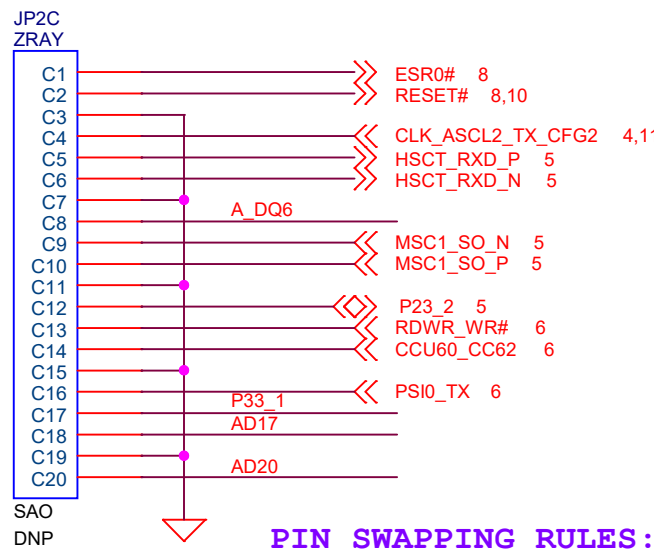
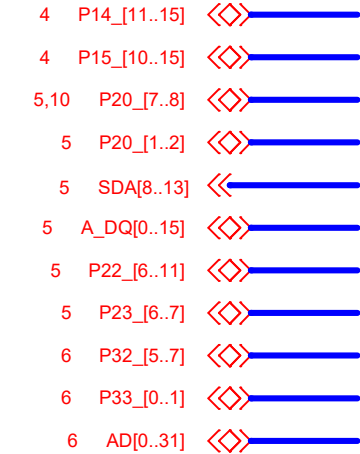
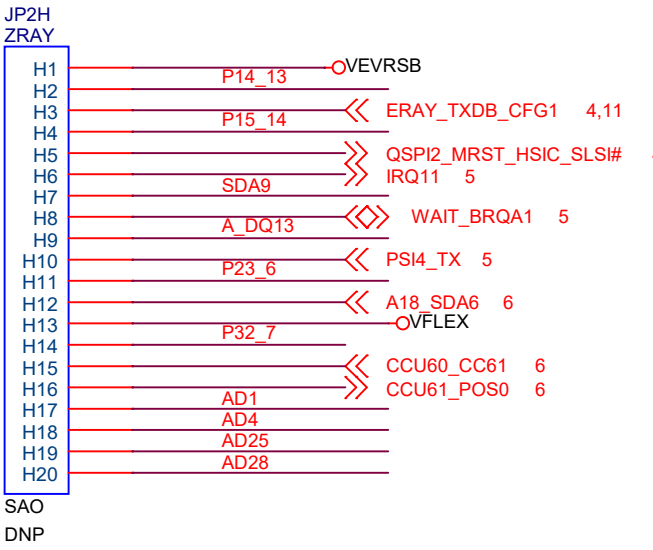
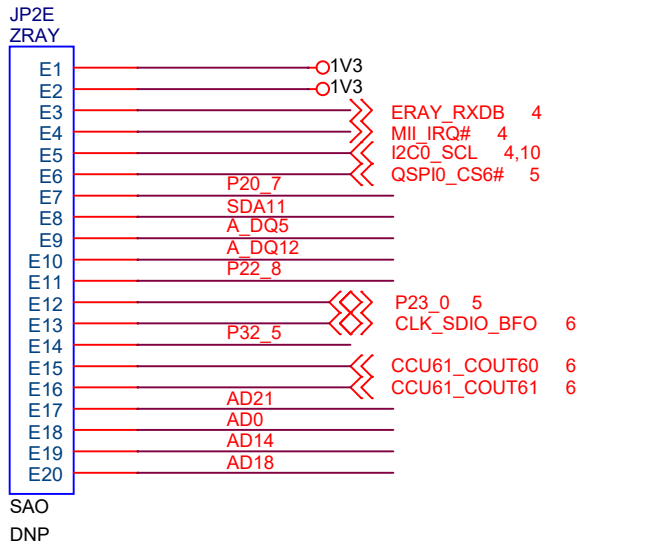
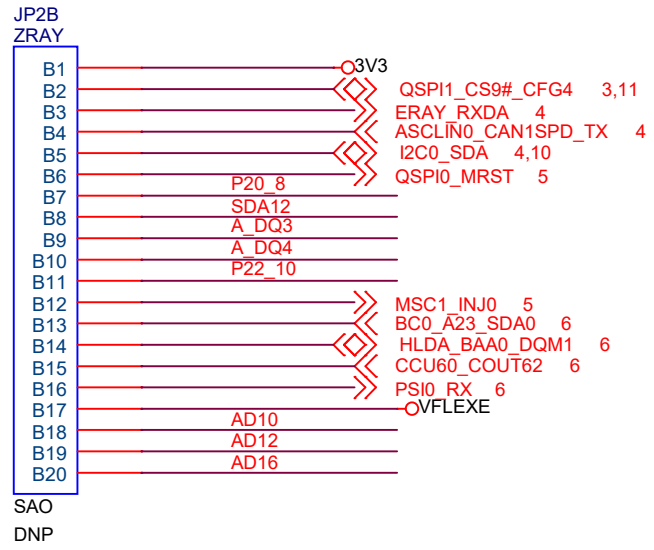
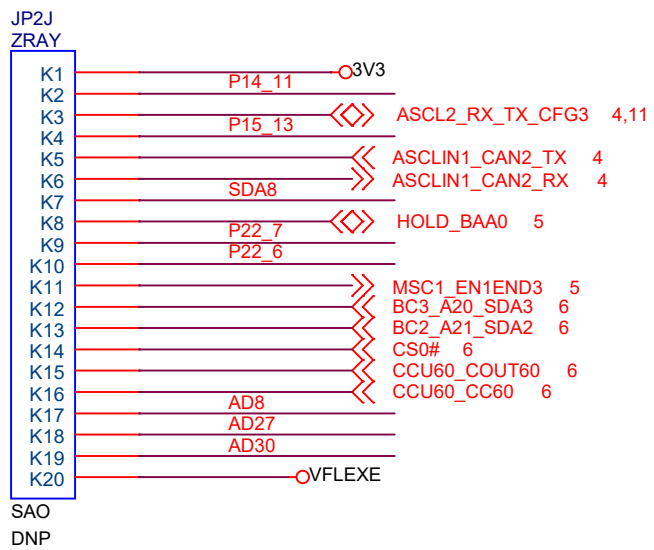
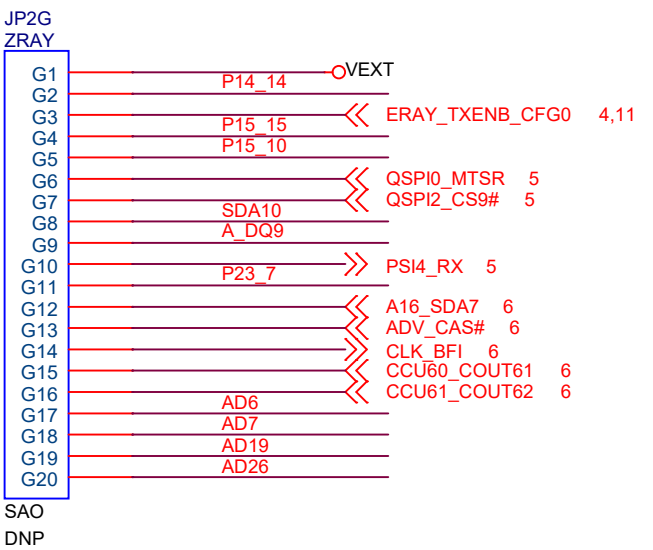
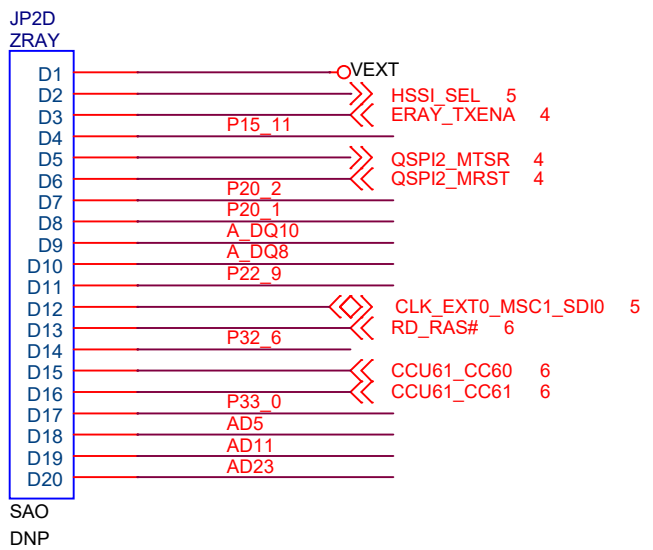
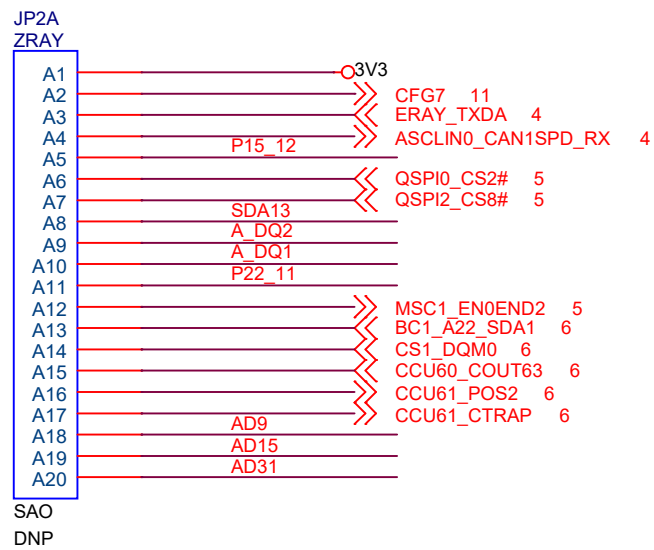
DEFAULT:
1, 2, 7 - "ON"
3, 4, 5, 6, 8 - "OFF"





PIN SWAPPING RULES:

- ANIx AND P40.x BETWEEN EACH OTHER - ANY WAY
- THE REST OF DIGITAL, EXCEPT CLOCK - ANY WAY
- CLOCK OR SIG_GND - ONLY IF NEEDED, KEEP GND_SIG AND CLOCK NEXT TO EACH OTHER
- POWER - ONLY IF NEEDED WITHIN ANALOG OR DIGITAL SECTION, KEEP GND_SIG/GND_A AND POWER NEXT TO EACH OTHER
- KEEP VAREF_x AND GND_VA_x NEXT TO EACH OTHER



PIN SWAPPING RULES:

- CLOCK, DIFF PAIRS OR SIG_GND - ONLY IF NEEDED, KEEP GND_SIG AND CLOCK/DIFF PAIRS NEXT TO EACH OTHER, DO NOT BREAK DIFF PAIRS
- ALL OF DIGITAL, EXCEPT CLOCKS AND DIFF PAIRS - ANY WAY
- POWER - ONLY IF NEEDED, KEEP GND_SIG AND POWER NEXT TO EACH OTHER

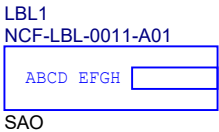
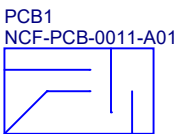
- All signals ending with _P/_N, except as noted, are differential pairs:
- shall be routed with 0.1mm width, 0.3mm edge-to-edge spacing on all the layers
 - length matched to better than 0.5mm between _P and _N
 - with >0.5mm clearance to other signals, >1mm clearance to board edge

TC297 pinout is a subset of TC299 but MAPPED OUT DIFFERNTLY - refer to TC297 datasheet for actual pin numbers

EXAMPLE: P00.0 on TC299 is M6, the same P00.0 on TC297 - in the same physical location on BGA array - is G1

TC39X pinout follow TC29x functionality as a bridge design - refer to TC39x datasheet for actual pin functions

TC39X 1.3V supply is actually 1.25V but kept as 1.3V notion to avoid re-routing planes



SILKSCREEN:

- NEUTRON CONTROLS LOGO
- TEXT "NCF AURIX CPU MODULE TC399/397 GEN1 REV A01"
- WHITEOUT SPACE

Title NCF AURIX TC399/397 MCU Module Gen1		
Size B	Document Number NCF_TC397_MCU_GEN1	Rev A01
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